AI-Based Smart Queue Prediction System

Python Program

This basic program estimates waiting time by multiplying the number of people in the queue by the average service time per person. It also classifies the queue as Low, Moderate, or High.

```
# AI-Based Smart Queue Prediction System

print("SMART QUEUE PREDICTION SYSTEM")
print("--------------------------------")

# Get input from the user
people = int(input("Enter the number of people in the queue: "))
service_time = float(input("Enter average service time per person (minutes): "))

# Predict waiting time
waiting_time = people * service_time

# Predict queue status
if people <= 5:
    queue_status = "Low Queue"
elif people <= 15:
    queue_status = "Moderate Queue"
else:
    queue_status = "High Queue"

# Display results
print("\n----- QUEUE PREDICTION RESULT -----")
print("Number of People:", people)
print("Average Service Time:", service_time, "minutes")
print("Predicted Waiting Time:", waiting_time, "minutes")
print("Queue Status:", queue_status)

print("--------------------------------")
print("Thank you for using the system!")
```

Sample Input

```
Enter the number of people in the queue: 10
Enter average service time per person (minutes): 3
```

Sample Output

```
SMART QUEUE PREDICTION SYSTEM
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----- QUEUE PREDICTION RESULT -----
Number of People: 10
Average Service Time: 3.0 minutes
Predicted Waiting Time: 30.0 minutes
Queue Status: Moderate Queue
--------------------------------
Thank you for using the system!
```

Note: This is a basic formula-based program, not a trained machine-learning model.